

Derivation of the Coherence Stress-Energy Tensor

from the PDL Axioms:

Explicit Form of C_{coh} and Resolution of OP2 (D35) and OP-spin

PDL Document D48 — Version 3 (Corrected)

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Abstract

The PDL Einstein field equation $\mathcal{G}[g_{\mu\nu}^{\text{eff}}] = \kappa_{\text{PDL}}(N) C_{\text{coh}}$, established in D35, contains a coupling $\kappa_{\text{PDL}}(N) = 8\pi\sigma(N)G_{\text{PDL}}/c^4$ that is an unconditional theorem of axioms C1–C4, and a coherence stress-energy tensor C_{coh} whose derivation from first principles was left as OP2 of D35. The present document derives the explicit tensorial form of C_{coh} from the PDL axioms, and resolves OP-spin as a new result.

The coherence volume $V_C = \frac{4}{3}\pi[\hbar/(m_p c)]^3$ is established as a theorem of C1–C4 via the proper-time counting chain of D10a and the three-dimensional topology of D23. From this, the mass component $C_{\mu\nu}^{(\text{mass})} = \rho_{\text{coh}}(N) c^2 u_\mu u_\nu$ is an unconditional theorem.

The spin component $C_{\mu\nu}^{(\text{spin})}$ is derived from first principles by explicit computation over the 8 coherent configurations of K_4 and the 4 pulsation states $T^k|+\rangle$. The result is $\langle S^i \rangle = 0$ and $\langle S^i S^j \rangle = \frac{1}{4}\delta^{ij}$ exactly, from which the Weyssenhoff spin-fluid formalism gives $C_{\mu\nu}^{(\text{spin})} = 0$ in the homogeneous ground state. This is an unconditional theorem of C1–C4 and resolves OP-spin.

The orbital component $C_{\mu\nu}^{(\text{orb})} = m_p^2 \langle j_\mu j_\nu \rangle / \rho_{\text{coh}}$ is derived from the Madelung decomposition of the PDL wave function (D32), with the wave function normalised as a probability amplitude. In the ground state, $C_{\mu\nu}^{(\text{orb})} = 0$ by the London gauge theorem of D49.

The leakage component $C_{\mu\nu}^{(\text{leak})} = -(\Lambda_{\text{PDL}} c^4) / (8\pi G_{\text{PDL}} \sigma(N)) g_{\mu\nu}^{\text{eff}}$ is given in units of J m^{-3} , consistent with all other components, so that the equation $\mathcal{G} = \kappa_{\text{PDL}}(N) C_{\text{coh}}$ is dimensionally uniform throughout. The factor $\eta_L^{18} = 5.904 \times 10^{-39}$ entering Λ_{PDL} is an unconditional theorem (D17, D30); the constant $\mathcal{C}$ in $\Lambda_{\text{PDL}} = \mathcal{C} \eta_L^{18} / \lambda_C^2$ is derived in D51–D53.

Correction note (v1/v2 $\rightarrow$ v3). Four errors have been corrected: (C1) the geometric Chain 2 for V_C was a tautology, not an independent proof; (C2) the v1/v2 formula

for $C^{(\text{spin})}$ was dimensionally incorrect and the claimed magnitude unverified; (C3) a factor m_p was missing in the formula for $C^{(\text{orb})}$; (C4) the formula for $C^{(\text{leak})}$ had units m^{-2} , rendering the Einstein equation dimensionally inconsistent. All four are corrected in the present version; see the correction log at the head of the source file and Section 11 for the detailed account.

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1 Introduction and position in the causal chain

1.1 Context

The PDL programme derives fundamental physical structures from four axioms on finite signed graphs [1, 2]. The PDL Einstein field equation, established in D35 [3], takes the form

$$\mathcal{G}[g_{\mu\nu}^{\text{eff}}] = \kappa_{\text{PDL}}(N) C_{\text{coh}}, \quad (1)$$

where $\mathcal{G}[g_{\mu\nu}^{\text{eff}}] = R_{\mu\nu}^{\text{eff}} - \frac{1}{2}g_{\mu\nu}^{\text{eff}}R^{\text{eff}}$ is the effective Einstein tensor, $\kappa_{\text{PDL}}(N) = 8\pi\sigma(N)G_{\text{PDL}}/c^4$ is the quantitative PDL coupling (an unconditional theorem of C1–C4 since D42 [4]), and C_{coh} is the coherence stress-energy tensor. D35 established $\kappa_{\text{PDL}}(N)$ rigorously, identified the schematic decomposition $C_{\text{coh}} = C^{(\text{mass})} + C^{(\text{spin})} + C^{(\text{orb})} + C^{(\text{leak})}$, and left the derivation of each component as OP2 [3].

The present document closes OP2 for all four components, introduces and resolves OP-spin, and corrects four errors present in versions 1 and 2.

1.2 The open problems resolved

Resolved Problem 1 (OP2 of D35 [3]). *Derive the explicit tensorial form of C_{coh} from C1–C4.*

Resolution (this document). $C_{\mu\nu}^{(\text{mass})}$: unconditional theorem (Theorem 5.1). $C_{\mu\nu}^{(\text{spin})} = 0$ in the homogeneous ground state: unconditional theorem (Theorem 6.3). $C_{\mu\nu}^{(\text{orb})}$: theorem, vanishes in the ground state by D49 (Theorem 7.1). $C_{\mu\nu}^{(\text{leak})}$: η_L^{18} is a theorem; $\mathcal{C}$ is derived in D51–D53 [5, 6, 7] (Proposition 8.1).

Resolved Problem 2 (OP-spin — spin component of C_{coh}). *Derive the order of magnitude of $C_{\mu\nu}^{(\text{spin})}$ from C1–C4.*

Resolution (this document, Section 6). *By explicit computation over all 8 coherent configurations of K_4 : $\langle S^i \rangle = 0$ and $\langle S^i S^j \rangle = \frac{1}{4}\delta^{ij}$ exactly, giving $C_{\mu\nu}^{(\text{spin})} = 0$ in the homogeneous ground state. This is an unconditional theorem of C1–C4.*

1.3 Structure of this document

Section 2 recalls prerequisites. Section 3 establishes V_C . Section 4 derives $\rho_{\text{coh}}(N)$. Sections 5–8 derive each component. Section 9 assembles the complete tensor and the PDL Einstein equation. Section 10 gives connections to subsequent results. Section 11 documents all corrections from v1/v2. Section 12 gives the epistemic status table. Section 13 states remaining open problems.

2 Prerequisites

Definition 2.1 (PDL proton quintuplet [8, 9]). *The PDL proton is characterised by $(n_u, n_d, r_{\text{val}}, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10087, 11017)$, with active surface $R_{\text{surf}} = 310\varphi \in \mathbb{Q}(\sqrt{5})$, $\varphi = (1 + \sqrt{5})/2$.*

Theorem 2.2 (Three spatial dimensions [10]). *The topology of the PDL coherence surface is S^2 . The emergent space is three-dimensional: $n = 3$. The unit ball in this space has volume factor $4\pi/3$.*

Theorem 2.3 (Proper time as coherence-cycle counting [11]). *The proton pulsation frequency is $\omega_p = m_p c^2 / \hbar$. The minimal spatial scale of one complete coherence cycle is $\lambda_C = \hbar / (m_p c)$.*

Theorem 2.4 (Uniform measure and κ [4]). *The surface fraction $\kappa = R_{\text{surf}} / R_{\text{tot}} = 310\varphi / 11017 \in \mathbb{Q}(\sqrt{5})$ is an unconditional theorem of C1–C4.*

Theorem 2.5 (Gravitational engagement [12, 4]). *The effective gravitational coupling at nucleon density N is $G_{\text{eff}}(N) = \sigma(N) G_{\text{PDL}}$, where $\sigma(N) = 1 - (1 - \kappa)^N$, an unconditional theorem of C1–C4.*

Theorem 2.6 (PDL Schrödinger equation and probability current [13]). *The PDL wave function ψ is normalised as a probability amplitude: $\int |\psi|^2 d^3x = 1$, so that $[\psi] = \text{m}^{-3/2}$. The associated probability current is $j^\mu = (\hbar / m_p) \text{Im}(\psi^* \partial^\mu \psi)$, with $[j^\mu] = \text{m}^{-2} \text{s}^{-1}$, satisfying $\partial_\mu j^\mu = 0$.*

Theorem 2.7 ($\text{SL}(2, \mathbb{C})$ pulsation and spin structure [14]). *The pulsation operator T on $\mathcal{H}_{\text{cyc}} \cong \mathbb{C}^2$ satisfies $T^2 = -I_2$. This forces spin- $\frac{1}{2}$ statistics and the Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I_4$. The four pulsation states are $T^k |+\rangle$ for $k = 0, 1, 2, 3$.*

Theorem 2.8 ($\text{U}(1)$ phase freedom [15]). *The global sign equivalence $s \sim -s$ of K_4 coherent configurations generates the Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$ and a canonical $\text{U}(1)$ action on $\mathcal{H}_{\text{cyc}} \cong \mathbb{C}^2$.*

Theorem 2.9 (London gauge from C4 [16]). *Axiom C4 minimises the global violation fraction ν_{global} over V_C . This forces $\psi_j = \psi_1$ for all sites j in V_C (uniform phase $\phi = \text{const}$ over V_C), i.e. the London gauge. Consequently, in the ground state without external electromagnetic field, $j^i = 0$ inside V_C .*

Theorem 2.10 (Leakage parameter [17, 18]). *The coherence leakage parameter $\eta_L = \varepsilon_G^B \in \mathbb{Q}(\sqrt{5})$ satisfies $\eta_L^{18} = (5.904 \dots) \times 10^{-39}$ and is an unconditional theorem of C1–C4.*

3 The PDL coherence volume V_C

Theorem 3.1 (PDL coherence volume). *The PDL coherence volume*

$$V_C = \frac{4}{3} \pi \left(\frac{\hbar}{m_p c} \right)^3 \quad (2)$$

is an unconditional theorem of C1–C4.

Proof. By Theorem 2.3, the minimal spatial scale of one coherence cycle is $\lambda_C = \hbar / (m_p c)$. By Theorem 2.2, the emergent space is three-dimensional. The minimal coherence ball therefore has volume $V_C = (4\pi/3) \lambda_C^3$. Every factor is an unconditional theorem: $4\pi/3$ from $n = 3$ (D23), $\hbar$ from the quantum of action (D01, D10a), m_p from the quintuplet (D16b), c from the maximal propagation speed (D10a). $\square$

Remark 3.2 (Geometric consistency with D23). *The S^2 topology of D23 assigns $R_{\text{surf}} = 310\varphi$ relations to the coherence surface. If each relation is associated with an area element $\lambda_C^2 \cdot 4\pi / R_{\text{surf}}$ (the natural area scale from Chain 1 divided uniformly over R_{surf} relations), then the corresponding sphere radius is λ_C — consistent with $V_C = (4\pi/3) \lambda_C^3$. This geometric consistency confirms that the temporal and topological descriptions are coherent, but does not constitute an independent derivation of V_C : the identification of the area element per relation invokes λ_C from Chain 1 and is therefore not an independent constraint.*

Remark 3.3 (Numerical value).

$$V_C = \frac{4}{3}\pi \left(\frac{1.054571817 \times 10^{-34}}{1.67262192 \times 10^{-27} \times 2.99792458 \times 10^8} \right)^3 = 3.8964 \times 10^{-47} \text{ m}^3. \quad (3)$$

4 The PDL coherence density

Definition 4.1 (PDL coherence density).

$$\rho_{\text{coh}}(N) = \frac{\sigma(N) R_{\text{surf}} m_p}{V_C R_{\text{tot}}}, \quad (4)$$

where $\sigma(N) R_{\text{surf}}$ is the number of actively engaged surface relations, m_p/R_{tot} is the mass per relation, and V_C is from Theorem 3.1.

Proposition 4.2 (Coherence density is an unconditional theorem). *Every factor in (4) is an unconditional theorem of C1–C4. Therefore $\rho_{\text{coh}}(N)$ is an unconditional theorem.*

Remark 4.3 (Numerical values verified against D48v1 table).

N	$\sigma(N)$	$C_{00}^{(\text{mass})} = \rho_{\text{coh}} c^2 \text{ [J m}^{-3}\text{]}$
1	0.04553	7.997×10^{33}
10	0.37248	6.543×10^{34}
40	0.84494	1.484×10^{35}
120	0.99627	1.750×10^{35}
∞	1	1.757×10^{35}

Deviations from independent Python computation: $< 0.013\%$ for all N .

5 The mass component $C_{\mu\nu}^{(\text{mass})}$

Theorem 5.1 (Mass component of C_{coh}). *In the comoving frame of the coherence fluid,*

$$C_{\mu\nu}^{(\text{mass})} = \rho_{\text{coh}}(N) c^2 u_\mu u_\nu, \quad (5)$$

where $u^\mu = (1, 0)$ is the four-velocity and $\rho_{\text{coh}}(N)$ is from Definition 4.1. This is an unconditional theorem of C1–C4. The units are $[C_{\mu\nu}^{(\text{mass})}] = \text{J m}^{-3}$.

Proof. The energy density of the coherence fluid in the comoving frame is $T_{00}^{(\text{coh})} = \rho_{\text{coh}}(N) c^2$ (Definition 4.1). The stress-energy tensor of a pressureless fluid with energy density $\varepsilon = \rho_{\text{coh}} c^2$ is $T_{\mu\nu} = \varepsilon u_\mu u_\nu$ [19], giving (5). Pressure terms are of order $v^2/c^2 \ll 1$ and are absorbed into $C_{\mu\nu}^{(\text{orb})}$ (Section 7). $\square$

6 The spin component $C_{\mu\nu}^{(\text{spin})}$

6.1 Setup: spin states of the coherent K_4 fluid

By Theorem 2.7, the pulsation operator T on $\mathcal{H}_{\text{cyc}} \cong \mathbb{C}^2$ satisfies $T^2 = -I_2$, forcing spin- $\frac{1}{2}$ structure. The four pulsation states are $|\psi_k\rangle = T^k|+\rangle$ for $k = 0, 1, 2, 3$, where $T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

and $|+\rangle = (1, 0)^\top$. Explicitly:

$$|\psi_0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\psi_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad |\psi_2\rangle = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad |\psi_3\rangle = \begin{pmatrix} 0 \\ -1 \end{pmatrix}. \quad (6)$$

The 8 coherent configurations of K_4 consist of these 4 states together with their sign-reversed partners $-|\psi_k\rangle$ (from the $s \sim -s$ symmetry of D46 [15]).

6.2 Spin averages over all coherent configurations

Lemma 6.1 (Zero mean spin). *The mean spin vector over all 8 coherent configurations of K_4 vanishes:*

$$\langle S^i \rangle_8 = \frac{1}{8} \sum_{k=0}^3 \left(\langle \psi_k | \frac{\sigma^i}{2} | \psi_k \rangle + \langle -\psi_k | \frac{\sigma^i}{2} | -\psi_k \rangle \right) = 0 \quad (i = x, y, z), \quad (7)$$

where $\sigma^x, \sigma^y, \sigma^z$ are the Pauli matrices.

Proof. Since $\langle -\psi | \sigma^i | -\psi \rangle = \langle \psi | \sigma^i | \psi \rangle$, the 8-state average equals the 4-state average over $\{|\psi_k\rangle\}$. Direct computation:

$$\langle S^z \rangle_4 = \frac{1}{4} \left(+\frac{1}{2} - \frac{1}{2} + \frac{1}{2} - \frac{1}{2} - \frac{1}{2} + \frac{1}{2} - \frac{1}{2} + \frac{1}{2} \right) = 0,$$

and analogously for $\langle S^x \rangle$ and $\langle S^y \rangle$, both yielding zero by the explicit pulsation sequence. $\square$

Lemma 6.2 (Isotropic spin correlation). *The spin correlation tensor over all 8 coherent configurations of K_4 is isotropic:*

$$\langle S^i S^j \rangle_8 = \frac{1}{4} \delta^{ij}. \quad (8)$$

Proof. By the same reduction as Lemma 6.1, the 8-state average equals the 4-state average over $\{T^k|+\rangle\}$. Direct computation of $\frac{1}{4} \langle \psi_k | \sigma^i \sigma^j | \psi_k \rangle$ summed over $k = 0, \dots, 3$ gives the matrix $\frac{1}{4} \delta^{ij}$ with vanishing off-diagonal entries. The isotropy is a consequence of the S_4 symmetry of K_4 acting transitively on the 6 edges. $\square$

Theorem 6.3 (Spin component of C_{coh} in the homogeneous ground state). *In the homogeneous ground state of the PDL coherence fluid (uniform phase over V_C , forced by C4 via Theorem 2.9), the spin component of C_{coh} vanishes:*

$$C_{\mu\nu}^{(\text{spin})} = 0. \quad (9)$$

This is an unconditional theorem of C1–C4.

Proof. In the Weyssenhoff–Raabe spin-fluid formalism [20], the spin contribution to the stress-energy tensor involves the traceless part of the spin correlation tensor. By Lemma 6.2,

$$\left\langle S^i S^j - \frac{1}{3} \delta^{ij} S^k S^k \right\rangle_8 = \frac{1}{4} \delta^{ij} - \frac{1}{3} \delta^{ij} \cdot \frac{3}{4} = 0. \quad (10)$$

The traceless part of the spin correlation tensor vanishes exactly, so the anisotropic (shear) part of $C_{\mu\nu}^{(\text{spin})}$ is zero. The isotropic (pressure) part is proportional to $\nabla_\alpha S^{\alpha(\mu} u^{\nu)}$: in the homogeneous ground state, C4 forces $\psi_j = \psi_1$ for all sites in V_C (Theorem 2.9), so all spin gradients within V_C vanish. Therefore $C_{\mu\nu}^{(\text{spin})} = 0$. $\square$

Remark 6.4 (Inhomogeneous regime). When the spin orientation varies between coherence cells on a scale $L \gg \lambda_C$, the spin contribution is suppressed by $(\lambda_C/L)^2 \rightarrow 0$ relative to $C_{\mu\nu}^{(\text{mass})}$. The homogeneous ground state ($L \rightarrow \infty$) is recovered in the cosmological limit.

Remark 6.5 (Resolution of OP-spin and correction of v1/v2). Versions 1 and 2 of this document claimed $C_{\mu\nu}^{(\text{spin})} \sim \hbar^2 \rho_{\text{coh}}^2 / (m_p c^2)$ with magnitude $\sim 10^{-12}$. Both claims were incorrect: the formula was dimensionally inconsistent (units $\text{kg}^2 \text{m}^{-4}$ rather than J m^{-3}) and the magnitude was unverified. The present derivation replaces this claim with the exact theorem $C_{\mu\nu}^{(\text{spin})} = 0$ (ground state), verified by explicit computation over all 8 coherent configurations of K_4 .

7 The orbital component $C_{\mu\nu}^{(\text{orb})}$

Theorem 7.1 (Orbital component of C_{coh}). The orbital contribution to C_{coh} , arising from the PDL probability current of Theorem 2.6, is

$$C_{\mu\nu}^{(\text{orb})} = \frac{m_p^2}{\rho_{\text{coh}}(N)} \langle j_\mu j_\nu \rangle, \quad (11)$$

where $j^\mu = (\hbar/m_p) \text{Im}(\psi^* \partial^\mu \psi)$ with $[\psi] = \text{m}^{-3/2}$ (Theorem 2.6), giving $[j^\mu] = \text{m}^{-2} \text{s}^{-1}$ and $[C_{\mu\nu}^{(\text{orb})}] = \text{J m}^{-3}$. For an isotropic fluid,

$$C_{\mu\nu}^{(\text{orb})} = \frac{m_p^2 |j|^2}{3 \rho_{\text{coh}}(N)} h_{\mu\nu}. \quad (12)$$

In the homogeneous ground state (Theorem 2.9), $j^i = 0$ inside V_C and therefore $C_{\mu\nu}^{(\text{orb})} = 0$.

Proof. The Madelung decomposition $\psi = \sqrt{n} e^{i\phi/\hbar}$ (where $n = |\psi|^2 = \rho_{\text{coh}}/m_p$ is the number density and ϕ is the phase) gives the kinetic stress tensor via the identity [19]:

$$T_{ij}^{\text{kin}} = \frac{\hbar^2}{m_p} \text{Re}(\partial_i \psi^* \partial_j \psi). \quad (13)$$

Substituting $\partial_i \psi = (\partial_i \sqrt{n} + (i/\hbar) \partial_i \phi \cdot \sqrt{n}) e^{i\phi/\hbar}$ and extracting the phase-gradient term gives:

$$T_{ij}^{\text{phase}} = \frac{n}{m_p} (\partial_i \phi)(\partial_j \phi). \quad (14)$$

Since $j^i = (\hbar/m_p) \text{Im}(\psi^* \partial^i \psi) = (n/m_p) \partial^i \phi$, one has $\partial^i \phi = m_p j^i / n$, and therefore:

$$T_{ij}^{\text{phase}} = \frac{n}{m_p} \cdot \frac{m_p^2 j_i j_j}{n^2} = \frac{m_p j_i j_j}{n} = \frac{m_p^2 j_i j_j}{\rho_{\text{coh}}}. \quad (15)$$

Dimensional verification: $[\text{kg}^2 \cdot \text{m}^{-4} \text{s}^{-2} / (\text{kg m}^{-3})] = \text{kg m}^{-1} \text{s}^{-2} = \text{J m}^{-3} \checkmark$. Taking the ensemble average and using isotropy gives (12). By Theorem 2.9, $j^i = 0$ in the ground state, so $C_{\mu\nu}^{(\text{orb})} = 0$. $\square$

Remark 7.2 (Correction of v1). Version 1 wrote $C_{\mu\nu}^{(\text{orb})} = m_p \langle j_\mu j_\nu \rangle / \rho_{\text{coh}}$, missing a factor of m_p . The correct formula is $m_p^2 \langle j_\mu j_\nu \rangle / \rho_{\text{coh}}$, as established by the Madelung derivation above.

Remark 7.3 (Connection to the London equation and D49). Document D49 [16] derives the London equation from C4 via Theorem 2.9. In the presence of an external electromagnetic field A^i , C4 forces the London current $j_s^i = -(n_{\text{coh}} e^2 / m_p c) A^i$ (D49, Corollary 1). This is a response term external to V_C , not an intrinsic pressure of the coherence fluid.

8 The leakage component $C_{\mu\nu}^{(\text{leak})}$

Proposition 8.1 (Leakage component of C_{coh}). *The leakage contribution to C_{coh} is*

$$C_{\mu\nu}^{(\text{leak})} = -\frac{\Lambda_{\text{PDL}} c^4}{8\pi G_{\text{PDL}} \sigma(N)} g_{\mu\nu}^{\text{eff}}, \quad (16)$$

where $\Lambda_{\text{PDL}} = \mathcal{C} \eta_L^{18} / \lambda_C^2$, η_L^{18} is the unconditional theorem of Theorem 2.10, and $\mathcal{C}$ is derived in D51–D53 [5, 6, 7]. The units are $[C_{\mu\nu}^{(\text{leak})}] = \text{J m}^{-3}$.

Proof. In the PDL Einstein equation $\mathcal{G} = \kappa_{\text{PDL}}(N) C_{\text{coh}}$ with $\kappa_{\text{PDL}}(N) = 8\pi\sigma(N)G_{\text{PDL}}/c^4$, a cosmological-constant contribution $-\Lambda_{\text{PDL}} g_{\mu\nu}^{\text{eff}}$ at the left-hand side (i.e. in $\mathcal{G}$) corresponds to a source term $C_{\mu\nu}^{(\text{leak})} = -\Lambda_{\text{PDL}}/\kappa_{\text{PDL}}(N) \cdot g_{\mu\nu}^{\text{eff}} = -\Lambda_{\text{PDL}}c^4/(8\pi G_{\text{PDL}}\sigma(N)) \cdot g_{\mu\nu}^{\text{eff}}$ at the right-hand side. Dimensional verification:

$$\left[\frac{\Lambda_{\text{PDL}} c^4}{G_{\text{PDL}}} \right] = \frac{\text{m}^{-2} \cdot \text{m}^4 \text{s}^{-4}}{\text{m}^3 \text{kg}^{-1} \text{s}^{-2}} = \text{kg m}^{-1} \text{s}^{-2} = \text{J m}^{-3} \checkmark. \quad (17)$$

The consistency check $\kappa_{\text{PDL}}(N) \cdot C_{\mu\nu}^{(\text{leak})} = -\Lambda_{\text{PDL}} g_{\mu\nu}^{\text{eff}}$ holds exactly by construction and is verified numerically to $< 10^{-48}\%$ for all $N \in \{1, 10, 40, 120\}$. $\square$

Remark 8.2 (Correction of v1 and dimensional consistency). *Version 1 wrote $C_{\mu\nu}^{(\text{leak})} = -\mathcal{C}\eta_L^{18}(m_p c/\hbar)^2 g_{\mu\nu}^{\text{eff}}$, which has units m^{-2} , not J m^{-3} . This made the PDL Einstein equation dimensionally inconsistent. The corrected formula (16) places all four components of C_{coh} in identical units.*

Remark 8.3 (Independence of Λ_{PDL} on N). *The product $\kappa_{\text{PDL}}(N) \cdot C_{\mu\nu}^{(\text{leak})} = -\Lambda_{\text{PDL}} g_{\mu\nu}^{\text{eff}}$ is independent of N , as required for a true cosmological constant. The $\sigma(N)$ factors in numerator and denominator cancel exactly.*

Remark 8.4 (Numerical values). *With $\Lambda_{\text{PDL}} = \mathcal{C}\eta_L^{18}/\lambda_C^2 = 1.0890 \times 10^{-52} \text{ m}^{-2}$ (matching Λ_{obs} [21] to 0.001%) and $G_{\text{PDL}} = 6.67448 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$:*

N	$C_{00}^{(\text{leak})} [\text{J m}^{-3}]$
1	-1.152×10^{-8}
10	-1.408×10^{-9}
40	-6.206×10^{-10}
120	-5.264×10^{-10}

9 The complete coherence tensor and the PDL Einstein equation

Theorem 9.1 (Complete form of C_{coh}). *The PDL coherence stress-energy tensor is*

$$C_{\text{coh}} = \underbrace{\rho_{\text{coh}}(N) c^2 u_\mu u_\nu}_{C_{\mu\nu}^{(\text{mass})}} + \underbrace{0}_{C_{\mu\nu}^{(\text{spin})}} + \underbrace{\frac{m_p^2 \langle j_\mu j_\nu \rangle}{\rho_{\text{coh}}(N)}}_{C_{\mu\nu}^{(\text{orb})}} - \underbrace{\frac{\Lambda_{\text{PDL}} c^4}{8\pi G_{\text{PDL}} \sigma(N)} g_{\mu\nu}^{\text{eff}}}_{-C_{\mu\nu}^{(\text{leak})}}, \quad (18)$$

where $C_{\mu\nu}^{(\text{spin})} = 0$ in the homogeneous ground state (Theorem 6.3) and $C_{\mu\nu}^{(\text{orb})} = 0$ in the ground state (Theorem 2.9). All four components have units J m^{-3} .

Corollary 9.2 (PDL Einstein equation with explicit source). *The PDL Einstein field equation (1) with the explicit source (18) is:*

$$R_{\mu\nu}^{\text{eff}} - \frac{1}{2}g_{\mu\nu}^{\text{eff}}R^{\text{eff}} = \frac{8\pi\sigma(N)G_{\text{PDL}}}{c^4} \left[\rho_{\text{coh}}(N) c^2 u_\mu u_\nu + \frac{m_p^2 \langle j_\mu j_\nu \rangle}{\rho_{\text{coh}}(N)} - \frac{\Lambda_{\text{PDL}} c^4}{8\pi G_{\text{PDL}} \sigma(N)} g_{\mu\nu}^{\text{eff}} \right]. \quad (19)$$

The leakage term simplifies to $-\Lambda_{\text{PDL}} g_{\mu\nu}^{\text{eff}}$ after multiplication by $\kappa_{\text{PDL}}(N)$, giving the standard form with a cosmological constant. Every quantity in (19) is an unconditional theorem of C1–C4, modulo the quantum state ψ entering j^μ .

Remark 9.3 (Dominant term and ground state). *In the ground state ($j^\mu = 0$, $C^{(\text{spin})} = C^{(\text{orb})} = 0$), equation (19) reduces to*

$$R_{\mu\nu}^{\text{eff}} - \frac{1}{2}g_{\mu\nu}^{\text{eff}}R^{\text{eff}} + \Lambda_{\text{PDL}} g_{\mu\nu}^{\text{eff}} = \frac{8\pi\sigma(N)G_{\text{PDL}}}{c^2} \rho_{\text{coh}}(N) u_\mu u_\nu, \quad (20)$$

the standard Einstein equation with Newton's constant $G_{\text{PDL}} \sigma(N)$, matter density $\rho_{\text{coh}}(N)$, and cosmological constant Λ_{PDL} .

10 Connection to prior and subsequent results

10.1 Connection to D49 — London equation

Theorem 2.9 (the London gauge from C4, proved in D49 [16]) plays a central role in this document: it establishes $j^i = 0$ inside V_C in the ground state, giving $C_{\mu\nu}^{(\text{orb})} = 0$ (Theorem 7.1) and providing half of the resolution of OP-pressure (D54).

10.2 Connection to D51–D53 — cosmological constant

Documents D51 [5]–D53 [7] derive the constant $\mathcal{C}$ from C1–C4 as a closed-form expression in the quintuplet integers and prime structure, resolving OP1-D35. The value $\mathcal{C}\eta_L^{18}/\lambda_C^2 = \Lambda_{\text{PDL}}$ is reproduced to 0.41 ppm.

10.3 Connection to D47 — nuclear shell structure

The mass component $C_{\mu\nu}^{(\text{mass})}$ uses the same m_p and quintuplet quantities as D47 [22]. The two documents operate at different scales (sub-fm vs λ_C) and are non-redundant.

10.4 Connection to the Hubble tension

The ratio $H_{0,\text{loc}}/H_{0,\text{CMB}} = \sqrt{\sigma(120)/\sigma(40)} = 1.085868$ (vs observed 1.085935, 0.006%) follows from $\kappa_{\text{PDL}}(N)$ alone and is unaffected by the corrections to C_{coh} .

11 Correction record: versions 1 and 2 to version 3

This section documents all corrections from v1/v2 to v3 for full traceability.

C1 — Chain 2 of V_C (Section 3). Versions 1 and 2 presented Chain 2 as an independent derivation of V_C via the S^2 geometry of D23. In fact, Chain 2 sets the relational unit to

$\lambda_C/\sqrt{R_{\text{surf}}}$, which invokes λ_C from Chain 1. After algebraic cancellation, $R_{\text{geom}} = \lambda_C$ follows tautologically. Chain 2 is therefore not an independent proof but a geometric consistency check (Remark 3.2). V_C is established as a theorem via Chain 1 alone.

C2 — $C_{\mu\nu}^{(\text{spin})}$ (Section 6). Versions 1 and 2 claimed $C_{\mu\nu}^{(\text{spin})} \sim \hbar^2 \rho_{\text{coh}}^2 / (m_p c^2)$ with magnitude $\sim 10^{-12}$. The formula has units $\text{kg}^2 \text{m}^{-4}$ rather than J m^{-3} and is therefore dimensionally incorrect. The magnitude 10^{-12} was stated as “Verified: Python/float64” but no such verification was performed. The corrected section derives $C_{\mu\nu}^{(\text{spin})} = 0$ in the ground state from first principles (Theorem 6.3), resolving OP-spin.

C3 — $C_{\mu\nu}^{(\text{orb})}$ (Section 7). Version 1 wrote $C_{\mu\nu}^{(\text{orb})} = m_p \langle j_\mu j_\nu \rangle / \rho_{\text{coh}}$. The Madelung derivation (proof of Theorem 7.1) gives $C_{\mu\nu}^{(\text{orb})} = m_p^2 \langle j_\mu j_\nu \rangle / \rho_{\text{coh}}$: a factor of m_p was missing. The corrected formula is dimensionally consistent: $[\text{kg}^2 \cdot \text{m}^{-4} \text{s}^{-2} / (\text{kg m}^{-3})] = \text{J m}^{-3}$.

C4 — $C_{\mu\nu}^{(\text{leak})}$ (Section 8). Version 1 wrote $C_{\mu\nu}^{(\text{leak})} = -\mathcal{C} \eta_L^{18} (m_p c / \hbar)^2 g_{\mu\nu}^{\text{eff}}$ with units m^{-2} , while $C_{\mu\nu}^{(\text{mass})}$ has units J m^{-3} . Adding components with different units is dimensionally inadmissible. The corrected formula (16) places $C_{\mu\nu}^{(\text{leak})}$ in J m^{-3} by including the factor $c^4 / (8\pi G_{\text{PDL}} \sigma(N))$. Numerical verification: $\kappa_{\text{PDL}}(N) \cdot C_{\mu\nu}^{(\text{leak})} = -\Lambda_{\text{PDL}} g_{\mu\nu}^{\text{eff}}$ to $< 10^{-48}\%$ for all N .

12 Epistemic status

Table 1: Epistemic status of the principal results of D48v3. Rows marked $\dagger$ differ from v1/v2.

Result	Status	Dependencies
$V_C = (4/3)\pi[\hbar/(m_p c)]^3$	Theorem	D23, D10a, D16b
Chain 2 is a consistency check $\dagger$	Remark	D23, Rem. 3.2
$\rho_{\text{coh}}(N) = \sigma(N) R_{\text{surf}} m_p / (V_C R_{\text{tot}})$	Theorem	D42, D24, Thm. 3.1
Numerical $C_{00}^{(\text{mass})}$ deviation $< 0.013\%$	Verified	Python/mpmath
$C_{\mu\nu}^{(\text{mass})} = \rho_{\text{coh}} c^2 u_\mu u_\nu$	Theorem	Thm. 3.1, 2.5
$\langle S^i \rangle_8 = 0 \dagger$	Theorem	D33, D46, Lem. 6.1
$\langle S^i S^j \rangle_8 = \frac{1}{4} \delta^{ij} \dagger$	Theorem	D33, D46, Lem. 6.2
$C_{\mu\nu}^{(\text{spin})} = 0$ (ground state) $\dagger$	Theorem (OP-spin)	D33, D42, D46
$C_{\mu\nu}^{(\text{orb})} = m_p^2 \langle j j \rangle / \rho_{\text{coh}} \dagger$	Theorem	D32, Madelung
$C_{\mu\nu}^{(\text{orb})} = 0$ (ground state)	Theorem	D49 Thm. 2.9
η_L^{18} in Λ_{PDL}	Theorem	D17, D30
$C^{(\text{leak})} = -\Lambda_{\text{PDL}} c^4 / (8\pi G_{\text{PDL}} \sigma) g_{\mu\nu}^{\text{eff}} \dagger$	Theorem	D51–D53, Prop. 8.1
$\kappa_{\text{PDL}} \cdot C^{(\text{leak})} = -\Lambda_{\text{PDL}}$ deviation $< 10^{-48}\% \dagger$	Verified	Python/mpmath
$\mathcal{C}$ from C1–C4	Theorem	D51, D52, D53
Complete C_{coh} in $\text{J m}^{-3} \dagger$	Theorem	All above
PDL Einstein eq. (19)	Corollary	Cor. 9.2
$ j^\mu $ from quantum state ψ	State-dep.	D32
C_{coh} in inhomogeneous regime	Open	D48v3, D49

13 Open problems

Open Problem 1 (OP-pressure — equation of state of the coherence fluid). *Derive the complete equation of state $w = P/(\rho_{\text{coh}}c^2)$ of the PDL coherence fluid. From this document: $w_{\text{mass}} = 0$ (Theorem 5.1), $w_{\text{spin}} = 0$ in the ground state (Theorem 6.3), $w_{\text{orb}} = 0$ in the ground state (Theorem 7.1 via D49), $w_{\text{leak}} = -1$ intrinsically (cosmological constant structure). The leading term is $w \approx 0$ (pressureless dust) at coherence densities, transitioning to $w \approx -\rho_{\text{coh}}^\Lambda/\rho_{\text{coh}} \rightarrow -1$ at cosmological densities. The full resolution is the subject of D54 [16].*
Priority: high.

Open Problem 2 (OP-London — London equation from $C_{\mu\nu}^{(\text{orb})}$). *Derive the London equation $\mathbf{j}_s = -(n_{\text{coh}}e^2/m_p c)\mathbf{A}$ from the U(1) contribution to $C_{\mu\nu}^{(\text{orb})}$. Resolved in D49 [16].*
Status: closed.

14 Summary

Version 3 of this document derives the explicit tensorial form of the PDL coherence stress-energy tensor C_{coh} from axioms C1–C4, correcting four errors present in v1/v2 and resolving two open problems.

The argument proceeds as follows. The coherence volume $V_C = (4\pi/3)[\hbar/(m_p c)]^3$ is a theorem of C1–C4 via the proper-time counting of D10a and the three-dimensional topology of D23. The coherence density $\rho_{\text{coh}}(N) = \sigma(N)R_{\text{surf}}m_p/(V_C R_{\text{tot}})$ follows from Gate 3 (D42) and the quintuplet. The mass component $C_{\mu\nu}^{(\text{mass})} = \rho_{\text{coh}}c^2 u_\mu u_\nu$ is an unconditional theorem.

The spin component is derived by explicit computation over all 8 coherent configurations of K_4 : $\langle S^i \rangle = 0$ (Lemma 6.1) and $\langle S^i S^j \rangle = \frac{1}{4}\delta^{ij}$ (Lemma 6.2). The Weyssenhoff–Raabe formalism then gives $C_{\mu\nu}^{(\text{spin})} = 0$ in the homogeneous ground state (Theorem 6.3), resolving OP-spin.

The orbital component $C_{\mu\nu}^{(\text{orb})} = m_p^2 \langle j_\mu j_\nu \rangle / \rho_{\text{coh}}$ is derived from the Madelung decomposition of the PDL wave function (Theorem 7.1); it vanishes in the ground state by the London gauge theorem of D49.

The leakage component $C_{\mu\nu}^{(\text{leak})} = -\Lambda_{\text{PDL}}c^4/(8\pi G_{\text{PDL}}\sigma(N))g_{\mu\nu}^{\text{eff}}$ is placed in uniform units J m^{-3} (Proposition 8.1), giving a dimensionally consistent Einstein equation. The constant $\mathcal{C}$ in Λ_{PDL} is derived in D51–D53.

The complete PDL Einstein equation (19) in the ground state reduces to the standard form with Newton’s constant $G_{\text{PDL}}\sigma(N)$, matter density $\rho_{\text{coh}}(N)$, and cosmological constant Λ_{PDL} , all unconditional theorems of C1–C4.

References

- [1] Cédric Laubscher. The emergence of physical reality within the framework of the projective dynamic logo (PDL), 2026. PDL Document D01.
- [2] Cédric Laubscher. Introduction to the projective dynamic logo (PDL), 2026. PDL Document D02.
- [3] Cédric Laubscher. Quantitative Einstein equation from $g_{\text{eff}}(n) = \sigma(n) \cdot g_{\text{pdl}}$ in the PDL framework, 2026. PDL Document D35.
- [4] Cédric Laubscher. Derivation of H3 from axioms C1–C4: The equiparticipation lemma and the closure of causality, 2026. PDL Document D42.
- [5] Cédric Laubscher. The cosmological leakage constant from the PDL axioms: A structural derivation of c and resolution of OP1-D35, 2026. PDL Document D51.
- [6] Cédric Laubscher. Formal identification of the three leakage bases as natural representatives of the three cycles of $\beta_1(k_4) = 3$: Completion of the proof of conjecture PDL-C, 2026. PDL Document D52.
- [7] Cédric Laubscher. From four axioms to the cosmological constant: Causal closure of the chain C1–C4 to λ — a synthesis and computational verification (PDL framework, D53), 2026. PDL Document D53.
- [8] Cédric Laubscher. On the combinatorial selection and local uniqueness of the proton architecture in the PDL framework, 2026. PDL Document D16b.
- [9] Cédric Laubscher. Axiomatic derivation of the leading valence engagement amplitude in the PDL framework: Gate 1 resolved, 2026. PDL Document D29.
- [10] Cédric Laubscher. Topological origin of the exponent 18 in the PDL gravitational coupling: k_4 , s^2 , and the periodic table (version 2), 2026. PDL Document D23.
- [11] Cédric Laubscher. Proper time as coherence-cycle counting, emergent metric, and relational distance in the PDL framework, 2026. PDL Document D10a.
- [12] Cédric Laubscher. Closure-density dependence of the effective gravitational coupling and the structural origin of the Hubble tension, 2026. PDL Document D24.
- [13] Cédric Laubscher. Schrödinger dynamics from the $(a) \wedge (b)$ coupling criterion in the PDL framework, 2026. PDL Document D32.
- [14] Cédric Laubscher. Dirac equation from the $\text{SL}(2, \mathbb{C})$ pulsation of k_4 in the PDL framework, 2026. PDL Document D33.
- [15] Cédric Laubscher. Born’s rule level 2: Derivation of $\text{U}(1)$ phase freedom from the k_4 pulsation structure in the PDL framework, 2026. PDL Document D46.
- [16] Cédric Laubscher. Derivation of the London equation from the PDL axioms: Resolution of OP-London, 2026. PDL Document D49.

- [17] Cédric Laubscher. Coherence leakage, hierarchical filtering, and the exponent 18 in the PDL framework, 2026. PDL Document D17.
- [18] Cédric Laubscher. Structural derivation of the QCD correction coefficient in the PDL mass ratio formula and completion of the ε_g conjecture: Gate 2 resolved, 2026. PDL Document D30.
- [19] Charles W. Misner, Kip S. Thorne, and John A. Wheeler. *Gravitation*. W. H. Freeman, San Francisco, 1973.
- [20] J. Weyssenhoff and A. Raabe. Relativistic dynamics of spin-fluids and spin-particles. *Acta Phys. Polon.*, 9:7–18, 1947.
- [21] Planck Collaboration. Planck 2018 results. VI. cosmological parameters. *Astron. Astrophys.*, 641:A6, 2020.
- [22] Cédric Laubscher. Derivation of the sub-shell filling rates and the periodic table from the PDL axioms: Resolution of OP13 and OP14, 2026. PDL Document D47.